Do Foundation Models Transfer Better Across Biomes? DINOv2/3 vs. U-Net for Forest Segmentation

C. S. Calderón Barrial[1]†, A. M. Vargas Zamora[1]†, E. Urbiola Ríos[1]†, F. Schwanebeck[1]†

[1]Pontificia Universidad Católica del Perú, Lima, Peru. †These authors contributed equally.
Corresponding author: c.calderonb@pucp.edu.pe

1. INTRODUCTION

Forest/non-forest segmentation of Sentinel-2 images is close to saturated inside one region: U-Nets reach an IoU of about 0.95 on the Amazon [2], [3], which leaves no room to tell encoders apart. Transfer to another biome is much harder: trained on the Amazon, a U-Net reaches a weighted IoU of only ≈ 0.81 on the Atlantic Forest [3]. Self-supervised foundation models such as DINOv2 [4] and DINOv3 [5], which also has a variant pretrained on satellite images, promise more general features. We ask: *do DINOv2 and DINOv3 encoders transfer better between biomes than an ImageNet-pretrained ResNet-34 U-Net, and does satellite pretraining help?* Our contribution is a controlled backbone swap evaluated as a cross-biome stress test.

2. METHODOLOGY

Data. We use the Amazon and Atlantic Forest dataset of Bragagnolo et al. [1]: Sentinel-2 L2A images of 512 × 512 px at 10 m with bands R, G, B, NIR and forest masks (Amazon 499 / 100 / 20 train / validation / test images, Atlantic Forest 485 / 100 / 20). The biomes share no Sentinel-2 tile.

Protocol. Each model is trained on one biome and tested on the official test split of both biomes and on the whole other biome (605 or 619 images), which it has never seen. Normalisation and checkpoint selection (best validation mIoU) use the training biome only.

Models. All encoders feed one U-Net decoder [6]; ViT features from four depths form a UNETR-style pyramid [7], and the first layer is adapted to four bands. Encoders: ResNet-34 (ImageNet), DINOv2 ViT-S/14, DINOv3 ViT-S/16, and DINOv3 ViT-L/16 pretrained on web (LVD-1689M) or satellite images (SAT-493M, 0.6 m RGB). Full fine-tuning (FT) trains all weights; partial fine-tuning (PFT) trains the first layer, the last ViT blocks (ResNet: stage 4) and the decoder. ViT-L is PFT only.

Training and statistics. Identical for all models: 40 epochs of random 256 px crops with flips and rotations, AdamW with a cosine schedule, BCE + Dice loss, one RTX 4090. Each model is trained with 5 seeds; a seed fixes the data order and augmentations for all models, so we report paired differences with 95 % CIs and Holm-adjusted p-values. Main metric: mIoU (mean of the forest and non-forest IoU).

3. RESULTS AND DISCUSSION

Table 1. mIoU, mean ± SD over 5 seeds. In = test split of the training biome; test = official test split of the other biome (20 images); whole = all images of the other biome (605 Atlantic Forest, 619 Amazon). Last row: weighted IoU, our range / John & Zhang [3].

Model	Amazon in	Amazon → Atlantic test	Amazon → Atlantic whole	Atlantic in	Atlantic → Amazon test	Atlantic → Amazon whole
U-Net ResNet-34, FT	0.964 ± 0.001	0.807 ± 0.009	0.814 ± 0.007	0.926 ± 0.002	0.918 ± 0.007	0.891 ± 0.011
U-Net ResNet-34, PFT	0.962 ± 0.002	0.814 ± 0.004	0.820 ± 0.003	0.929 ± 0.000	0.924 ± 0.006	0.905 ± 0.009
DINOv2 ViT-S/14, FT	0.960 ± 0.001	0.816 ± 0.003	0.823 ± 0.003	0.924 ± 0.001	0.933 ± 0.007	0.899 ± 0.012
DINOv2 ViT-S/14, PFT	0.958 ± 0.001	0.816 ± 0.007	0.824 ± 0.005	0.923 ± 0.001	0.938 ± 0.003	0.909 ± 0.009
DINOv3 ViT-S/16, FT	0.964 ± 0.001	0.816 ± 0.007	0.818 ± 0.007	0.931 ± 0.002	0.929 ± 0.005	0.901 ± 0.010
DINOv3 ViT-S/16, PFT	0.962 ± 0.002	0.811 ± 0.006	0.814 ± 0.005	0.930 ± 0.001	0.934 ± 0.006	0.907 ± 0.012
DINOv3 ViT-L web, PFT	0.965 ± 0.000	0.819 ± 0.011	0.822 ± 0.013	0.932 ± 0.001	0.923 ± 0.004	0.899 ± 0.008
DINOv3 ViT-L sat, PFT	0.965 ± 0.001	0.802 ± 0.013	0.806 ± 0.010	0.933 ± 0.000	0.921 ± 0.011	0.881 ± 0.018
Weighted IoU: ours / [3]	*0.958–0.965 / 0.947–0.952*	*0.823–0.840 / 0.813–0.814*	–	*0.933–0.942 / 0.888–0.920*	*0.918–0.938 / 0.825–0.872*	–

In-biome results are saturated (Table 1). All eight models reach 0.958–0.965 mIoU on the Amazon test split and 0.923–0.933 on the Atlantic Forest split. Yet 5 of the 20 Atlantic Forest test

images show the same ground as training images on another date, so these numbers are optimistic.

Transfer hits the same barrier for every encoder. Trained on the Amazon, all models drop to 0.802–0.819 mIoU on the Atlantic Forest test split (0.806–0.824 on all 605 images). In the weighted IoU of [3] we reach 0.82–0.84 against their ≈ 0.81. The errors are mostly false forest (on all 605 images precision 0.81–0.85, recall 0.90–0.91; forest area over-estimated by 2–4 points), and all models mark the same crop fields as forest (Fig. 1). The reverse direction transfers much better (0.88–0.91 on the whole Amazon), with the opposite error.

Encoders and pretraining. Relative to the U-Net (FT), no gain is significant after Holm correction over the seven comparisons (all adjusted $p > 0.28$); the largest are +0.010 mIoU on the whole biome (DINOv2 ViT-S PFT) and +0.012 on the test split (DINOv3 ViT-L web). Satellite pretraining does not help: DINOv3 ViT-L sat equals the web version in-biome (0.965) but is 0.017 lower on the Atlantic Forest test split (95 % CI [−0.041, +0.006], better in 1 of 5 seeds), while training 112 M of its 314 M parameters (U-Net: 24 M) for 6× the time. A higher in-biome score does not predict better transfer (Spearman $\rho = -0.31$ against whole-biome mIoU over the 40 Amazon-trained runs; −0.10 on the test split).

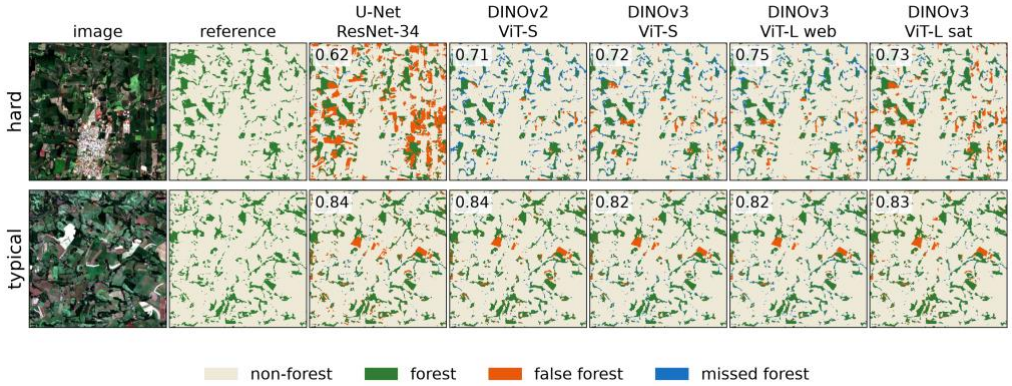


Fig. 1. Amazon-trained models on two Atlantic Forest test images (seed 0; hard and typical image = 10th and 50th percentile of the mean image mIoU; U-Net and ViT-S: FT; number = image mIoU). Orange = false forest, blue = missed forest.

4. CONCLUSIONS

Self-supervised foundation-model encoders gain at most ≈ 0.01 mIoU over an ImageNet U-Net when transferring from the Amazon to the Atlantic Forest: all eight variants, including a 314 M-parameter DINOv3 ViT-L pretrained on satellite images, stop at ≈ 0.81 mIoU, and no gain is significant. In-biome benchmarks on this data are saturated and partly leaky, so encoder comparisons need cross-biome tests; the shared errors suggest that the limit is the shift between the biomes (look-alike land cover, label definitions) rather than the features. **Limitations:** one dataset and one pair of biomes, 20 official test images per biome, five seeds, one training schedule, four bands. **Next steps:** domain adaptation, multi-biome training, a label audit and Sentinel-1. Code, result files and checkpoints: github.com/dsbferris/WAIMLAp-2026.

REFERENCES

[1] L. Bragagnolo, R. V. da Silva, and J. M. V. Grzybowski, “Amazon and Atlantic Forest image datasets for semantic segmentation,” Zenodo, 2021, doi: 10.5281/zenodo.4498086.
[2] L. Bragagnolo, R. V. da Silva, and J. M. V. Grzybowski, “Amazon forest cover change mapping based on semantic segmentation by U-Nets,” Ecol. Inform., vol. 62, p. 101279, 2021.
[3] D. John and C. Zhang, “An attention-based U-Net for detecting deforestation within satellite sensor imagery,” Int. J. Appl. Earth Obs. Geoinf., vol. 107, p. 102685, 2022.
[4] M. Oquab et al., “DINOv2: Learning robust visual features without supervision,” Trans. Mach. Learn. Res., 2024.
[5] O. Siméoni et al., “DINOv3,” arXiv:2508.10104, 2025.
[6] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in Proc. MICCAI, 2015, pp. 234–241.
[7] A. Hatamizadeh et al., “UNETR: Transformers for 3D medical image segmentation,” in Proc. IEEE/CVF WACV, 2022, pp. 574–584.